

Project Selection Methods

PRESENT WORTH METHOD

FUTURE WORTH METHOD

ANNUAL EQUIVALENT METHOD

What is Project

An investment may be thought of as **an exchange of resources now for an expected flow of benefits in the future**. A company may be able to use funds to install equipment that will reduce labour costs in the future. These funds might otherwise have been used on another project or returned to the shareholders or owners. An individual may be able to study to become an engineer. Studying requires that time be given up that could have been used to earn money or to travel. The benefit of study, though, is the expectation of a good income from an interesting job in the future.

Engineers play a major role in making decisions about investment opportunities. In many cases, they are the ones who estimate the expected costs of and returns from an investment. They then **must decide whether the expected returns outweigh the costs to see if the opportunity is potentially acceptable**. They may also have to examine **competing investment opportunities to see which is best**. Engineers frequently refer to investment opportunities as projects. Throughout most of this text, the term *project* will be used to mean *investment opportunity*.

Relation among projects

The simplest relation between projects occurs when they are **independent**. Two projects are independent if the expected costs and the expected benefits of each project do not depend on whether the other one is chosen.

When two or more projects are independent, evaluation is simple. Consider each opportunity one at a time, and accept or reject it on its own merits.

Projects are **mutually exclusive** if, in the process of choosing one, all other alternatives are excluded. In other words, two projects are mutually exclusive if it is impossible to do both or it clearly would not make sense to do both.

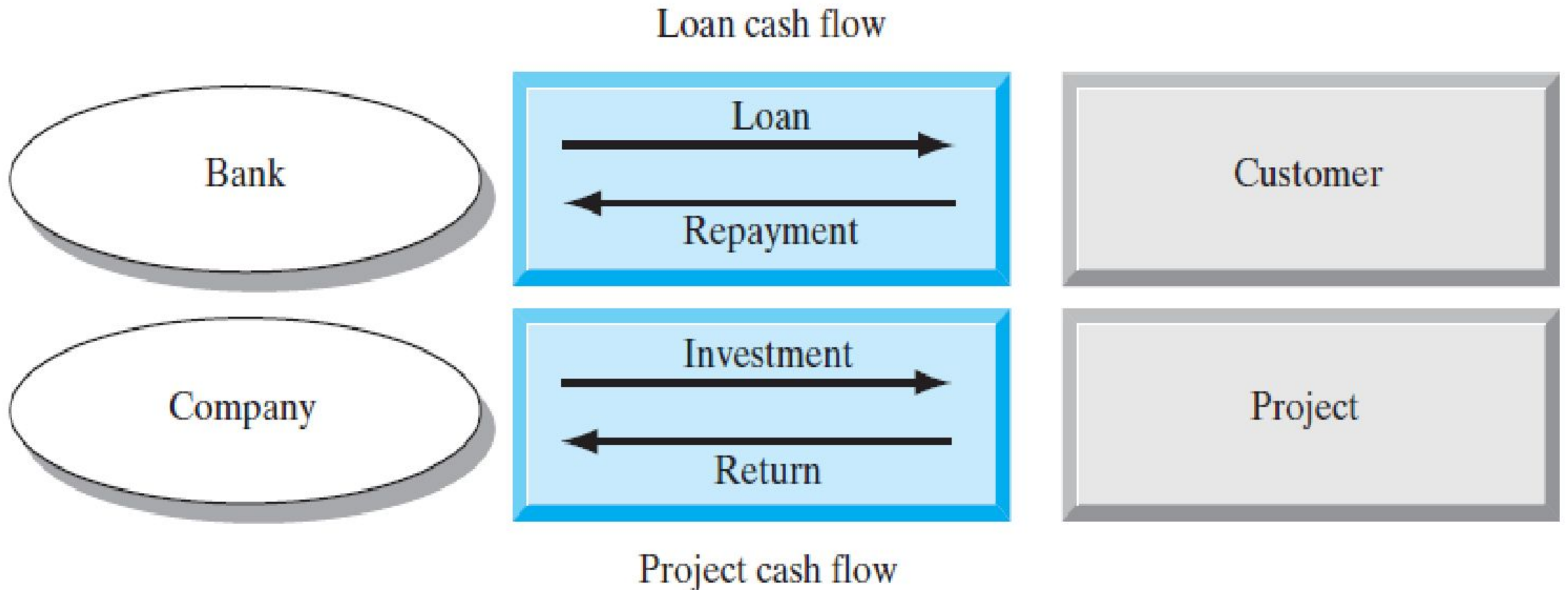
For example, consider a student about to invest in a computer printer. She can get an inkjet printer or a laser printer, but it would not make sense to get both. She would consider the options to be mutually

Loan versus Project Cash Flows

An investment made in a fixed asset is similar to an investment made by a bank when it lends money. The essential characteristic of both transactions is that funds are committed today in the expectation of their earning a return in the future. **In the case of the bank loan, the future return takes the form of interest plus repayment of the principal** and is known as the **loan cash flow**.

In the case of the fixed asset, **the future return takes the form of cash generated by productive use of the asset**. As shown figure, the representation of these future earnings, along with the capital expenditures and annual expenses (such as wages, raw materials, operating costs, maintenance costs, and income taxes), is the **project cash flow**.

Loan versus Project Cash Flows



Minimum Acceptable Rate of Return (MARR)

The MARR is an interest rate that must be earned for any project to be accepted. Projects that earn at least the MARR are desirable, since this means that the money is earning at least as much as can be earned elsewhere. Projects that earn less than the MARR are not desirable, since investing money in these projects denies the opportunity to use the money more profitably elsewhere. The MARR is thus an opportunity cost in two senses.

Present Worth Method-Steps

With the present worth method, the analyst compares project A and project B by computing the present worths of the two projects at the MARR. The preferred project is the one with the greater present worth.

One can also speak of *present cost*. That is, the best project is the one with the minimum present worth of cost as opposed to the maximum present worth.

Present Worth Method-Steps

In a cost dominated cash flow approach, the costs (outflows) will be assigned with positive sign and the profit, revenue, salvage value (all inflows), etc. will be assigned with negative sign.

In a revenue/profit-dominated cash flow diagram, the profit, revenue, salvage value (all inflows to an organization) will be assigned with positive sign. The costs (outflows) will be assigned with negative sign.

In case the decision is to select the alternative with the minimum cost, then the alternative with the least present worth amount will be selected. On the other hand, if the decision is to select the alternative with the maximum profit, then the alternative with the maximum present worth will be selected.

Decision Rule for Independent VS. Mutually Exclusive Projects

- The present worth of any money invested at the MARR is zero, since the present worth of future receipts would exactly offset the current disbursement. Consequently, if an independent project has a present worth greater than zero, it is acceptable. If an independent project has a present worth less than zero, it is unacceptable. If an independent project has a present worth of exactly zero, it is considered *marginally acceptable*.
- It is very easy to use the present worth method to choose the best project among a set of mutually exclusive projects *when the service lives are the same*. One just computes the present worth of each project using the MARR. The project with the greatest present worth is the preferred project because it is the one with the greatest profit.

Independent Project

Steve Chen, a third-year electrical engineering student, is really into gaming. He and all of his friends like online multi-player games, but are annoyed by delays caused by slow internet connections. Sometimes they link their own computers into local networks and play, but it's hard to find a place to play and their laptops are not the best machines for gaming. On his work term in Abu Dhabi, Steve noticed that gaming halls are popular. High-level networked game machines are located in malls, and people rent the machines by the hour for playing games either with each other or over a very fast internet connection. Steve expects to be on campus for the next five years and sees an opportunity in setting up a similar business at the mall next to his university. The first cost for equipment, furniture, and software is expected to be \$70 000. Students will be able to rent time on the computers by the hour, and Steve will be able to augment this income with some auxiliary services. Net annual cash flow, after paying for labour, supplies, and other costs, is expected to be \$30 000 a year for five years. When Steve finishes his graduate program in five years, he plans to close the business. The five-year-old equipment and furniture are expected to have zero value. If investors in this type of service enterprise demand a return of 20 percent per year, is this a good investment?

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$$\begin{aligned}PW &= -70\,000 + 30\,000(P/A, 20\%, 5) \\&= -70\,000 + 30\,000(2.9906) \\&= 19\,718 \\&\cong 20\,000\end{aligned}$$

-70,000 q kay woh pehly hi cost lag gyi, worth sai hatt gyi

30,000 usko hamesha laganey hain jisses twenty percent usko mileyga

yani investors ko ekar bhi which is 20% MARR usko faida hoga hi

The project is acceptable, since the present worth of about \$20 000 is greater than zero. it was an indepent project

A mechanical engineer is considering building automated materials-handling equipment for a production line. On the one hand, the equipment would substantially reduce the manual labour currently required to move items from one part of the production process to the next. On the other hand, the equipment would consume energy, require insurance, and need periodic maintenance.

Alternative 1: Continue to use the current method. Yearly labour costs are \$9200.

Alternative 2: Build automated materials-handling equipment with an expected service life of 10 years.

Assume MARR is 9% per y

First cost	\$15 000	
Labour	\$3300	per year
Power	\$400	per year
Maintenance	\$2400	per year
Taxes and insurance	\$300	per year

mutually exclusive since both cannot be done together

Alternative 1: Continue to use the current method. Yearly labour costs are \$9200.
Alternative 2: Build automated materials-handling equipment with an expected service life of 10 years.

First cost	\$15 000	
Labour	\$3 300	per year
Power	\$400	per year
Maintenance	\$2 400	per year
Taxes and insurance	\$300	per year

this table belongs to alternative 2

this is a revenue dominating solution, in which we calculated that what will be the benefit of choosing over alternative 2 over alternative 1

$$\begin{aligned}
 PW &= -15\,000 + [9200 - (3300 + 400 + 2400 + 300)](P/A, 9\%, 10) \\
 &= -15\,000 + 2800(P/A, 9\%, 10) \\
 &= -15\,000 + 2800(6.4176) \\
 &= 2969.44
 \end{aligned}$$

The present worth of the cost savings is approximately \$3000 greater than the \$15 000 first cost. Therefore, alternative 2 is worth implementing.

Alpha Industry is planning to expand its production operation. It has identified three different technologies for meeting the goal. The initial outlay and annual revenues with respect to each of the technologies are summarized in Table 4.1. Suggest the best technology which is to be implemented based on the present worth method of comparison assuming 20% interest rate, compounded annually.

Table 4.1

	<i>Initial outlay</i> (Rs.)	<i>Annual revenue</i> (Rs.)	<i>Life</i> (years)
Technology 1	12,00,000	4,00,000	10
Technology 2	20,00,000	6,00,000	10
Technology 3	18,00,000	5,00,000	10

The present worth expression for this technology is

20% interest compounded annually is treated same as ordinary

$$\begin{aligned}PW(20\%)_1 &= -12,00,000 + 4,00,000 \times (P/A, 20\%, 10) \\&= -12,00,000 + 4,00,000 \times (4.1925) \\&= -12,00,000 + 16,77,000 \\&= \text{Rs. } 4,77,000\end{aligned}$$

The present worth expression for this technology is

$$\begin{aligned}PW(20\%)_2 &= -20,00,000 + 6,00,000 \times (P/A, 20\%, 10) \\&= -20,00,000 + 6,00,000 \times (4.1925) \\&= -20,00,000 + 25,15,500 \\&= \text{Rs. } 5,15,500\end{aligned}$$

The present worth expression for this technology is

$$\begin{aligned}PW(20\%)_3 &= -18,00,000 + 5,00,000 \times (P/A, 20\%, 10) \\&= -18,00,000 + 5,00,000 \times (4.1925) \\&= -18,00,000 + 20,96,250 \\&= \text{Rs. } 2,96,250\end{aligned}$$

From the above calculations, it is clear that the present worth of technology 2 is the highest among all the technologies. Therefore, technology 2 is suggested for implementation to expand the production.

EXAMPLE 4.2 An engineer has two bids for an elevator to be installed in a new building. The details of the bids for the elevators are as follows:

<i>Bid</i>	<i>Engineer's estimates</i>		
	<i>Initial cost</i> (Rs.)	<i>Service life (years)</i>	<i>Annual operations & maintenance cost (Rs.)</i>
Alpha Elevator Inc.	4,50,000	15	27,000
Beta Elevator Inc.	5,40,000	15	28,500

Determine which bid should be accepted, based on the present worth method of comparison assuming 15% interest rate, compounded annually.

The total present worth cost of bid 1 is less than that of bid 2. Hence, bid 1 is to be selected for implementation. That is, the elevator from Alpha Elevator Inc. is to be purchased and installed in the new building.

cost dominated problem, so mn selected and cost taken as pos and rev as neg

$$\begin{aligned}PW(15\%) &= 4,50,000 + 27,000(P/A, 15\%, 15) \\&= 4,50,000 + 27,000 \times 5.8474 \\&= 4,50,000 + 1,57,879.80 \\&= \text{Rs. } 6,07,879.80\end{aligned}$$

$$\begin{aligned}PW(15\%) &= 5,40,000 + 28,500(P/A, 15\%, 15) \\&= 5,40,000 + 28,500 \times 5.8474 \\&= 5,40,000 + 1,66,650.90 \\&= \text{Rs. } 7,06,650.90\end{aligned}$$

EXAMPLE 4.3 Investment proposals A and B have the net cash flows as follows:



<i>Proposal</i>	<i>End of years</i>				
	0	1	2	3	4
A (Rs.)	-10,000	3,000	3,000	7,000	6,000
B (Rs.)	-10,000	6,000	6,000	3,000	3,000

Compare the present worth of A with that of B at $i = 18\%$. Which proposal should be selected?

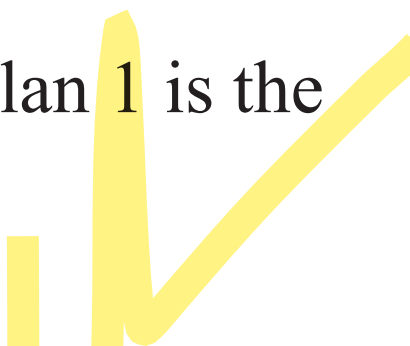
EXAMPLE 4.5 A finance company advertises two investment plans. In plan 1, the company pays Rs. 12,000 after 15 years for every Rs. 1,000 invested now. In plan 2, for every Rs. 1,000 invested, the company pays Rs. 4,000 at the end of the 10th year and Rs. 4,000 at the end of 15th year. Select the best investment plan from the investor's point of view at $i = 12\%$, compounded annually.

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$$\begin{aligned}PW(12\%) &= -1,000 + 12,000(P/F, 12\%, 15) \\&= -1,000 + 12,000(0.1827) \\&= \text{Rs. } 1,192.40\end{aligned}$$

$$\begin{aligned}PW(12\%) &= -1,000 + 4,000(P/F, 12\%, 10) + 4,000(P/F, 12\%, 15) \\&= -1,000 + 4,000(0.3220) + 4,000(0.1827) \\&= \text{Rs. } 1,018.80\end{aligned}$$

The present worth of plan 1 is more than that of plan 2. Therefore, plan 1 is the best plan from the investor's point of view.



EXAMPLE 4.4 A granite company is planning to buy a fully automated granite cutting machine. If it is purchased under down payment, the cost of the machine is Rs. 16,00,000. If it is purchased under installment basis, the company has to pay 25% of the cost at the time of purchase and the remaining amount in 10 annual equal installments of Rs. 2,00,000 each. Suggest the best alternative for the company using the present worth basis at $i = 18\%$, compounded annually.

$$\begin{aligned}PW(18\%) &= 4,00,000 + 2,00,000(P/A, 18\%, 10) \\&= 4,00,000 + 2,00,000 \times 4.4941 \\&= \text{Rs. } 12,98,820\end{aligned}$$

The present worth of this option is Rs. 12,98,820, which is less than the first option of complete down payment of Rs. 16,00,000. Hence, the company should select the second alternative to buy the fully automated granite cutting machine.

this was a cost dominated issue with cost +

EXAMPLE 4.6 Novel Investment Ltd. accepts Rs. 10,000 at the end of every year for 20 years and pays the investor Rs. 8,00,000 at the end of the 20th year. Innovative Investment Ltd. accepts Rs. 10,000 at the end of every year for 20 years and pays the investor Rs. 15,00,000 at the end of the 25th year. Which is the best investment alternative? Use present worth base with $i = 12\%$.



EXAMPLE 4.7 A small business with an initial outlay of Rs. 12,000 yields Rs. 10,000 during the first year of its operation and the yield increases by Rs. 1,000 from its second year of operation up to its 10th year of operation. At the end of the life of the business, the salvage value is zero. Find the present worth of the business by assuming an interest rate of 18%, compounded annually.



EXAMPLE 5.1 Consider the following two mutually exclusive alternatives:

<i>Alternative</i>	<i>End of year</i>				
	0	1	2	3	4
A (Rs.)	−50,00,000	20,00,000	20,00,000	20,00,000	20,00,000
B (Rs.)	−45,00,000	18,00,000	18,00,000	18,00,000	18,00,000

At $i = 18\%$, select the best alternative based on future worth method of comparison.

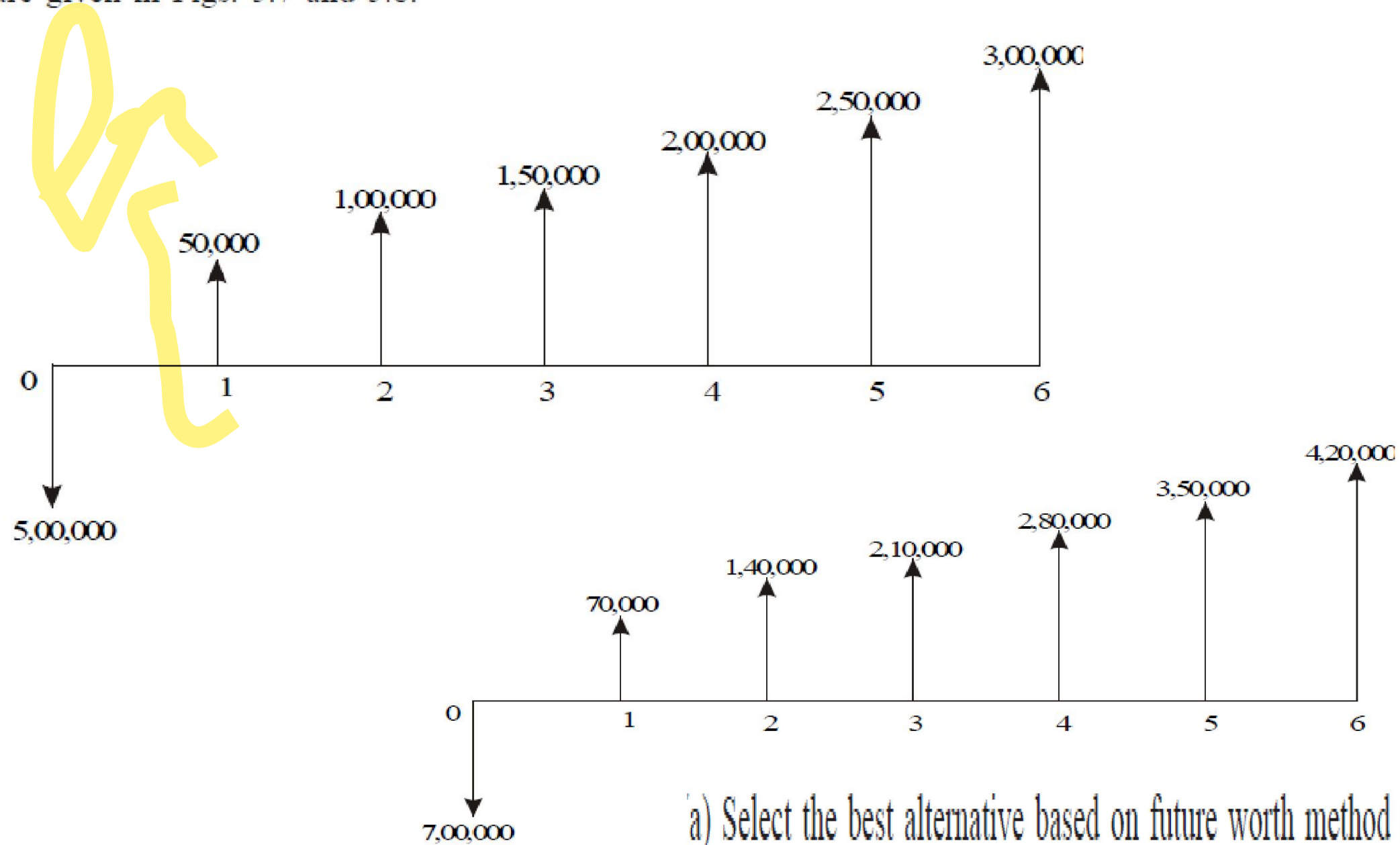


EXAMPLE 5.2 A man owns a corner plot. He must decide which of the several alternatives to select in trying to obtain a desirable return on his investment. After much study and calculation, he decides that the two best alternatives are as given in the following table:

	<i>Build gas station</i>	<i>Build soft ice-cream stand</i>
First cost (Rs.)	20,00,000	36,00,000
Annual property taxes (Rs.)	80,000	1,50,000
Annual income (Rs.)	8,00,000	9,80,000
Life of building (years)	20	20
Salvage value (Rs.)	0	0

Evaluate the alternatives based on the future worth method at $i = 12\%$.

EXAMPLE 5.3 The cash flow diagram of two mutually exclusive alternatives are given in Figs. 5.7 and 5.8.



a) Select the best alternative based on future worth method at $i = 8\%$

$$\begin{aligned}
 FW_1(8\%) &= -P(F/P, 8\%, 6) + [A1 + G(A/G, 8\%, 6)] \times (F/A, 8\%, 6) \\
 &= -5,00,000(1.587) + [50,000 + 50,000(2.2764)] \times 7.336 \\
 &= -79,35,000 + 1,63,820 \times 7.336 \\
 &= -79,35,000 + 12,01,784 \\
 &= \text{Rs. } 4,08,283.52
 \end{aligned}$$


$$\begin{aligned}
 FW_2(8\%) &= -P(F/P, 8\%, 6) + [A1 + G(A/G, 8\%, 6)] \times (F/A, 8\%, 6) \\
 FW_2(8\%) &= -7,00,000 \times 1.587 + [70,000 + 70,000 \times 2.2764] \times 7.336 \\
 &= -11,10,900 + 16,82,497 \\
 &= \text{Rs. } 5,71,596.93
 \end{aligned}$$


Annual Equivalent Method

In the annual equivalent method of comparison, first the annual equivalent cost or the revenue of each alternative will be computed. Then the alternative with the maximum annual equivalent revenue in the case of revenue-based comparison or with the minimum annual equivalent cost in the case of cost based comparison will be selected as the best alternative.

EXAMPLE 6.2 A company is planning to purchase an advanced machine centre. Three original manufacturers have responded to its tender whose particulars are tabulated as follows:

<i>Manufacturer</i>	<i>Down payment</i> (Rs.)	<i>Yearly equal installment</i> (Rs.)	<i>No. of installments</i>
1	5,00,000	2,00,000	15
2	4,00,000	3,00,000	15
3	6,00,000	1,50,000	15

Determine the best alternative based on the annual equivalent method by assuming $i = 20\%$, compounded annually.

Solution Alternative 1

Down payment, $P = \text{Rs. } 5,00,000$

Yearly equal installment, $A = \text{Rs. } 2,00,000$

$n = 15$ years

$i = 20\%$, compounded annually

$$\begin{aligned}
 AE_1(20\%) &= 5,00,000(A/P, 20\%, 15) + 2,00,000 \\
 &= 5,00,000(0.2139) + 2,00,000 \\
 &= 3,06,950
 \end{aligned}$$

Alternative 2

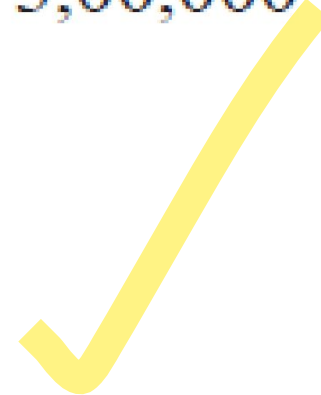
Down payment, $P = \text{Rs. } 4,00,000$

Yearly equal installment, $A = \text{Rs. } 3,00,000$

$n = 15$ years

$i = 20\%$, compounded annually

$$\begin{aligned} AE_2(20\%) &= 4,00,000(A/P, 20\%, 15) + 3,00,000 \\ &= 4,00,000(0.2139) + 3,00,000 \\ &= \text{Rs. } 3,85,560. \end{aligned}$$



Alternative 3

Down payment, $P = \text{Rs. } 6,00,000$

Yearly equal installment, $A = \text{Rs. } 1,50,000$

$n = 15$ years

$i = 20\%$, compounded annually



$$\begin{aligned} AE_3(20\%) &= 6,00,000(A/P, 20\%, 15) + 1,50,000 \\ &= 6,00,000(0.2139) + 1,50,000 \\ &= \text{Rs. } 2,78,340. \end{aligned}$$

The annual equivalent cost of manufacturer 3 is less than that of manufacturer 1 and manufacturer 2. Therefore, the company should buy the advanced machine centre from manufacturer 3.